ZT Tree Inventory

Tree Inventory Report

Tree inventory and GPS tree mapping (approximate, not a survey)

SAMPLE - Bristol Park (fake trees)

Client	Sample Engineering Co. (fake)
Site address	123 Sample St, Pensacola FL (fake)
Parcel ID(s)	03-1S-31-1500-001-003
Jurisdiction	Escambia County (unincorporated)
Field date	2026-10-22
Crew	ZT
Trees inventoried	6
Coordinate system	NAD83 / Florida North (ftUS) - EPSG:2238
Report date	10/22/2026
Prepared by	Zach Taylor, EI (Engineer Intern), ZT, LLC

Job notes: Home test-run style sample

LOCATIONS APPROXIMATE, FOR FIELD REFERENCE ONLY. NOT A SURVEY.

Tool, not engineering or surveying advice. Not survey grade and not a substitute for a licensed surveyor or engineer; the client's licensed surveyor locates the tags. Tree condition and risk determinations require an ISA Certified Arborist; condition entries here are visual field notes only. Prepared by Zach Taylor, EI (Engineer Intern), ZT, LLC.

Phone GPS is not survey grade: expect roughly 10-30 ft horizontal error (GPS.gov: smartphones are typically accurate to within 4.9 m / 16 ft under open sky, and accuracy worsens near buildings, bridges, and trees; https://www.gps.gov/gps-accuracy). The client's licensed surveyor locates the tags.

Contents: summary and protected-tree count & mitigation worksheet; tree table; location sketch; accuracy statement and field accuracy checks; data sources; limitations.

Summary

6 trees inventoried. Field calls: 3 protected, 1 heritage. Code-threshold suggestions (before exemptions, field judgement): 3 protected, 0 heritage. Total DBH 99.5 in.

Position quality: 4 within the 30 ft job limit, 1 flagged worse than 30 ft, 1 entered by hand (accuracy unknown).

Species	Count	Total DBH (in)	Protected (field)	Heritage (field)
Southern live oak	1	32.0	1	1
Longleaf pine	1	14.0	0	0
Laurel oak	1	22.5	1	0
Southern magnolia	1	11.0	0	0
Sweet bay	1	8.0	0	0
Water oak	1	12.0	1	0
TOTAL	6	99.5	3	1

Protected-tree count & mitigation worksheet

Counts to carry into the client's mitigation calculation; which trees are removed or preserved is decided by the client's design team, not recorded here. Inch-for-inch and replacement ratios differ by jurisdiction; the jurisdiction makes the final determination. Protected/heritage flags are suggested from the jurisdiction's published DBH thresholds and are field judgement only; the jurisdiction makes the final determination.

Item	Count	DBH (in)
All trees inventoried	6	99.5
Protected trees (field call)	3	66.5
Heritage trees (field call)	1	32.0
Protected by code DBH threshold (suggestion)	3	66.5
Heritage by code DBH threshold (suggestion)	0	0.0
Live trees	6	99.5
Dead trees (not counted by the calculator)	0	0.0

Jurisdiction rule basis for the suggestions: Escambia LDC / DSM Ch. 2 Art. 2. The tree log XLSX includes a "Calculator paste" sheet for the mitigation calculator.

Tree table

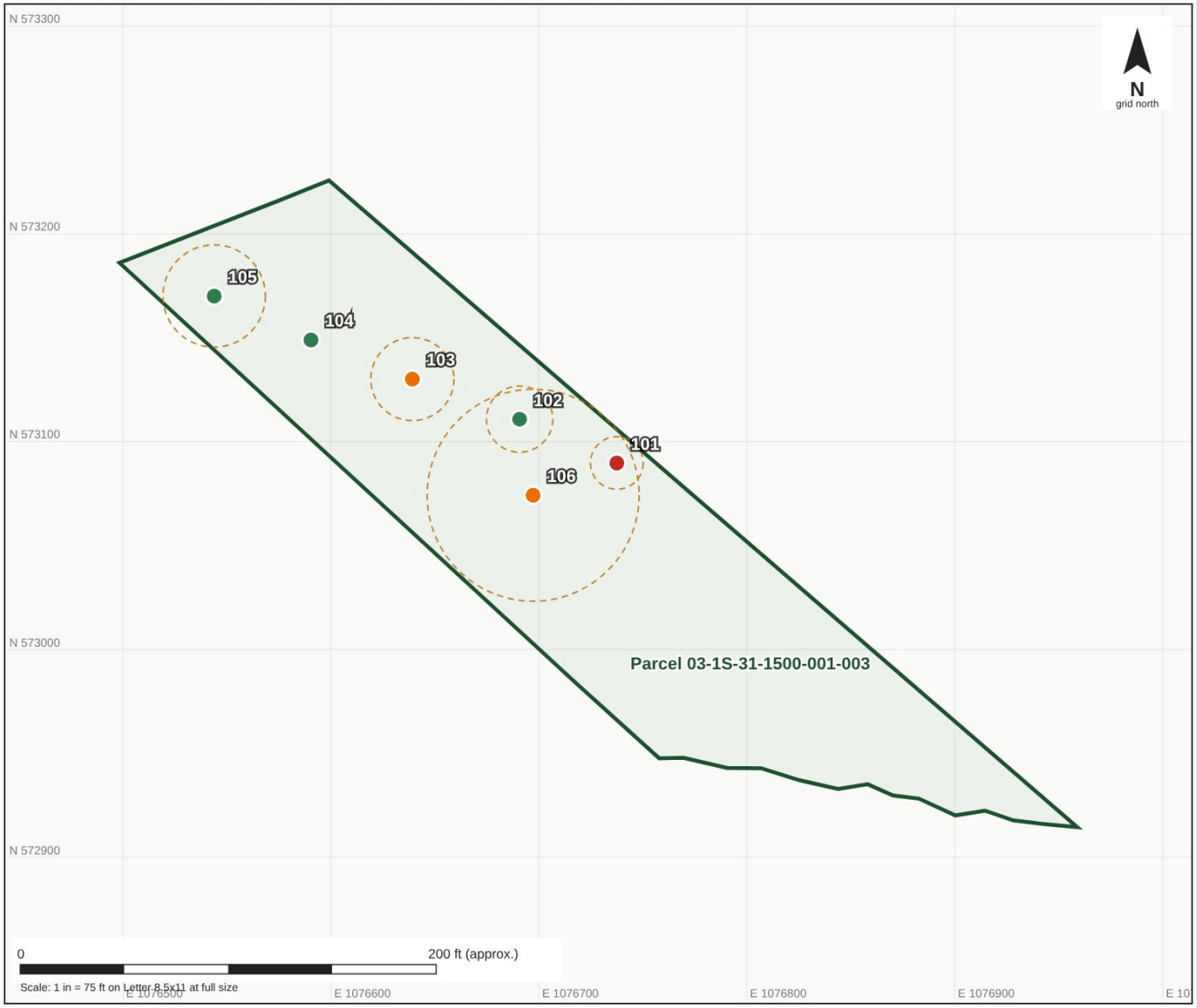
Condition is a visual field note only. P/H = protected/heritage field call; "Sugg." = code-threshold suggestion. N/E in NAD83 / Florida North (ftUS) - EPSG:2238. +/-ft = phone-reported accuracy.

Tag	Pt #	Species	DBH	Canopy ft	Condition (visual)	P	H	Sugg. P/H	+/-ft	Northing	Easting	Notes
101	1000	Southern live oak	32	30	Good	Y	Y	Y/N	13	573089.7	1076737.5	FAKE sample tree
102	1001	Longleaf pine	14	12	Good	N	N	N/N	16	573110.8	1076690.7	
103	1002	Laurel oak	22.5	22	Good	Y	N	Y/N	20	573130.1	1076639.1	
104	1003	Southern magnolia	11	10	Fair	N	N	N/N	manual	573148.9	1076590.4	
105	1004	Sweet bay	8	8	Good	N	N	-/-	25	573170.0	1076543.8	
106	1005	Water oak	12		Good	Y	N	Y/N	51 !	573074.2	1076697.2	[accuracy: dense canopy, re-shoot later]

Tree Location Sketch (approximate)

LOCATIONS APPROXIMATE, FOR FIELD REFERENCE ONLY. NOT A SURVEY.

Accuracy: Phone GPS is not survey grade: expect roughly 10-30 ft horizontal error (GPS.gov: smartphones are typically accurate to within 4.9 m / 16 ft under open sky, and accuracy worsens near buildings, bridges, and trees; https://www.gps.gov/gps-accuracy). The client's licensed surveyor locates the tags.



ZT Tree Inventory — tree inventory data

Job	SAMPLE - Bristol Park (fake trees)
Client	Sample Engineering Co. (fake)
Site	123 Sample St, Pensacola FL (fake)
Parcel(s)	03-1S-31-1500-001-003
Jurisdiction	Escambia County (unincorporated)
Field date	2026-10-22
Trees	6
CRS	NAD83 / Florida North (ftUS) - EPSG:2238
Prepared by	Zach Taylor, EI (Engineer Intern), ZT, LLC

Legend

- Tree (tag # label)
- Protected (field call, suggested by code DBH)
- Heritage (field call, suggested by code DBH)
- Parcel line (county GIS, approx., not survey)
- Reported GPS accuracy radius

Tag	Pt #	Species	DBH	±ft
101	1000	Southern live oak	32	13
102	1001	Longleaf pine	14	16
103	1002	Laurel oak	22.5	20
104	1003	Southern magnolia	11	man.
105	1004	Sweet bay	8	25
106	1005	Water oak	12	51

Coordinate system: NAD83 / Florida North (ftUS) - EPSG:2238. Positions are phone GPS readings (averaged where noted), projected with proj4js; no datum shift applied. Parcel lines: Escambia County Property Appraiser (parcels); source acknowledgment: Escambia County IMAGiNE GIS, retrieved 10/22/2026, 11:02:01 PM; parcel lines are approximate and not survey quality.
Background map: none.

Locations approximate, for field reference only. Not a survey. Tool, not engineering or surveying advice. Not survey grade and not a substitute for a licensed surveyor or engineer; the client's licensed surveyor locates the tags. Tree condition and risk determinations require an ISA Certified Arborist; condition entries here are visual field notes only. Prepared by Zach Taylor, EI (Engineer Intern), ZT, LLC.

Accuracy statement

Phone GPS is not survey grade: expect roughly 10-30 ft horizontal error (GPS.gov: smartphones are typically accurate to within 4.9 m / 16 ft under open sky, and accuracy worsens near buildings, bridges, and trees; https://www.gps.gov/gps-accuracy). The client's licensed surveyor locates the tags.

Method: each tree position is the weighted average (by 1/accuracy squared) of repeated phone GPS readings taken at the trunk; the "+/-ft" value is the phone's own estimate, not a measured error. Coordinates are converted from WGS84 latitude/longitude to NAD83 / Florida North (ftUS) - EPSG:2238 with proj4js; no datum shift is applied (about 1-2 m, well inside phone GPS error).

Source: GPS.gov, GPS Accuracy, https://www.gps.gov/gps-accuracy.

Field accuracy checks (known points)

The phone was placed on points with known coordinates and the horizontal offset measured. PASS = offset within 30 ft.

Point	Known source	Offset ft	dN	dE	Phone +/-ft	Readings	Result
NE corner iron rod (FAKE)	test fixture	18.2	18.2	0.5	15	6	PASS
Monument (FAKE)	test fixture	48.1	35.6	32.4	18	6	FAIL (> 30 ft)

Checks: 2; mean offset 33.2 ft; max 48.1 ft.

Data sources

• GPS accuracy: GPS.gov, "GPS Accuracy" (smartphones typically within 4.9 m / 16 ft under open sky; worse near buildings, bridges and trees) (https://www.gps.gov/gps-accuracy)

• Coordinate conversion: proj4js with EPSG definitions from epsg.io (default EPSG:2238 NAD83 / Florida North, US survey ft); spot-checked against NOAA NGS NCAT (https://geodesy.noaa.gov/NCAT/)

• Escambia County and City of Pensacola parcels: Escambia County Property Appraiser; source acknowledgment: Escambia County IMAGiNE GIS (not survey quality) (https://gismaps.myescambia.com/arcgis/rest/services/Individual_Layers/parcels/FeatureServer/0)

• Santa Rosa County parcels: Santa Rosa County BOCC GIS (accuracy, completeness and currency not guaranteed) (https://www.santarosa.fl.gov/470/Mapping-GIS/)

• Sketch background (when used): USGS The National Map imagery (public domain) (https://basemap.nationalmap.gov/)

• City of Pensacola protected/heritage DBH thresholds: Pensacola Land Development Code Ch. 12-6 and Appendix A (as encoded in the ZT tree mitigation calculator rules; confirm current code with the City)

• Escambia County protected/heritage thresholds: Escambia County LDC and Design Standards Manual Ch. 2, Art. 2 (confirm current code with the County)

• Santa Rosa County protected/heritage thresholds: Santa Rosa County LDC 4.07.05 (confirm current code with the County)

• Tree risk documentation requirement: Florida Statutes 163.045 (ISA Certified Arborist or Florida licensed landscape architect) (https://www.flsenate.gov/Laws/Statutes/2025/163.045)

Limitations

• This is tree inventory data and GPS tree mapping (approximate, not a survey). Locations approximate, for field reference only. Not a survey.

• Positions are phone GPS readings with roughly 10-30 ft horizontal error; the client's licensed surveyor locates the tags before any design or permit use that depends on location.

• Tool, not engineering or surveying advice. This is not engineering work, a boundary or tree location survey, landscape architecture, or an arborist report. The preparer is an Engineer Intern (EI), not a licensed professional engineer, surveyor, landscape architect or arborist.

• Tree condition and risk determinations require an ISA Certified Arborist; condition entries here are visual field notes only. No tree risk or hazard assessment was performed. Under F.S. 163.045, tree risk documentation must come from an ISA Certified Arborist or a Florida licensed landscape architect.

• Protected/heritage flags are suggested from the jurisdiction's published DBH thresholds and are field judgement only; the jurisdiction makes the final determination.

• Parcel lines are county GIS (property appraiser) data, not survey quality, and can be offset from true boundaries by tens of feet.

• Species identification and DBH are field observations. DBH was measured or estimated at 4.5 ft above grade; multi-trunk DBH uses the square root of the sum of squared stem DBHs (Pensacola LDC 12-6-2(e); Escambia/Santa Rosa method unverified).

• The mitigation worksheet is a count to support the client's own permit calculations; the jurisdiction makes final protected/heritage and mitigation determinations.

Prepared by Zach Taylor, EI (Engineer Intern), ZT, LLC. ZT Tree Inventory v1.0.1.